machine for a few cents. The person need not establish a bank account for this purpose; neither do they have to remember or keep track of the machine after cashing the collection. Similarly, in the decentralized channel, you do not want the worker to incur more overhead than the payment they may collect. Also, they should not be cashing the same micropayment repeatedly (double-spending, in Bitcoin terminology). These issues are precipitated by the fact that online participants in MPC are temporary and are typically unknown to the organization. That's why the smart contract is a transient channel between the organizer and a worker. Also, the cost of deploying and undeploying the MPC contract is significantly less than the typical micropayments to the workers.

These ideas about temporary channel are new concepts that you may not have seen in your traditional programming, but you may want to remember them when designing blockchain programs with smart contracts.

7.4.5 MPC application development (MPC-app)

Let's develop `MPC-app`, the off-chain part of the Dapp where the user interface is typically located. Recall from chapter 4 the design pattern or structure for the MPC-Dapp:

- A web application hosted on a Node.js server
- A web stack with a user interface
- app.js implementing the application's glue code with web3 calls for interfacing with the blockchain services

Among the off-chain operations--plastics collection, verification of bins at a station that informs the organizer, and digital signing